Timelines

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1. Lisa invest R15000 (initial amount) for five years. For the first three years it earns interest at 8.5% p.a. compounded quarterly, and for the last two years, 7.5% p.a compounded semi-annually. What is the total interest earned over five years? (three years; two years)

$A = P(1+i)^n$

$A_1 = 15000\left(1+\frac{0.085}{4}\right)^{12}$

$A_1 = R19305.27949$

$A_2 = 19305.27949\left(1+\frac{0.075}{2}\right)^{4}$

$A_2 = R22368.0701$

$A = R22368.0701 - R15000 = R7368.07$

2. A savings account was opened with an initial deposit of R24000, 18 months later R7000 was drawn from the account. Calculate how much money will be in the savings account at the end of 4 years if the interest rate was 10.5% pa compounded monthly. (~1.5 year; minus)

(1.5 years)

$A = P(1+i)^n$

$A = 24000\left(1+\frac{0.105}{12}\right)^{18}$

$A = R28074.69849 - R7000$

$A_1 = R21074.69849$

(2.5 years)

$A = P(1+i)^n$

$A = 21074.69849\left(1+\frac{0.105}{12}\right)^{30}$

$A = R27369.55913$

3. Bantu Biko made an initial deposit of R32000 into an investment account that paid interest at 8.6% p.a compounded monthly. Another deposit of R23000 was made 3 years later. The interest rate changed to 10.5% p.a compounded quarterly 4 years after the initial deposit.

3.1 How much money was in Bantu Biko's investment account at the end of 4 years?

3.2 At the end of 6 years since he started his investment, Bantu decided to use all his balance as a deposit for a car that costs R220 000 and borrow the rest from the bank. How much did he need to borrow?

3.1 $A_1 = P(1+i)^n$

$m \times n = 12 \times 4 = 48$

$A_1 = 32000\left(1+\frac{0.086}{12}\right)^{48}$

$A_1 = 45083.17396$

12 x 1

$A_2 = 23000\left(1+\frac{0.086}{12}\right)^{12}$

$A_2 = R25057.85907$

4 years ~ 3 years

4 − 3 = 1

$A = A_1 + A_2 = R45083.17396 + R25057.85907 = R70141.03303$

(4 years)

3.2 $A = P(1+i)^n$

(4 years)

3.2 $A = P(1+i)^n$

$A = 70141.03303\left(1+\frac{0.105}{4}\right)^8$

$A = R86297.36175$

$R220000 - R86297.36175 = R133702.6382$

Bantu Biko needs to borrow R133702.6382 from the bank.

4. Hlumelo deposited R25000 into a savings account with an interest rate of 18% p.a. compounded monthly. ~m=12
Hlumelo withdrew R8000 from the account 2 years after* depositing the initial amount. She deposited another R4000 into this account $3\frac{1}{2}$ (~3.5) years after the initial deposit. What amount will Hlumelo have 5 years after making the initial deposit into this account?

$A = P(1+i)^n$

$A = 25000\left(1+\frac{0.18}{12}\right)^{24}$

$A = 35737.5703 - 8000$

$A = R27737.5703$ (money after 2 years)

* $A_1 = 27737.5703\left(1+\frac{0.18}{12}\right)^{36}$

$A_1 = R47407.37809$

(1.5 years)

$A_2 = 4000\left(1+\frac{0.18}{12}\right)^{18}$

$A_2 = R5229.362543$

$A = A_1 + A_2$

$A = R47407.37809 + R5229.362543$

$A = R52636.74063$